
Education

- 2021/9 - 2026/4 **Doctor of Philosophy**, Earth Sciences, Simon Fraser University
Thesis: [Gaussian process emulation of subglacial drainage for model calibration, uncertainty quantification and efficient coupled ice-sheet modelling](#)
Supervisors: Dr. Gwenn Flowers, Dr. Derek Bingham
- 2019/9 - 2021/8 **Master of Mathematics**, Applied Mathematics, University of Waterloo
Thesis: [Mathematical modelling of supraglacial meltwater production and drainage](#)
Supervisor: Dr. Christine Dow
- 2014/9 - 2019/4 **Bachelor of Science**, Honours Co-operative Mathematical Physics, University of Waterloo

Appointments

- 2026/9 - present **Research Associate**, Center for Climatic Research, University of Wisconsin-Madison
- Coupled modelling of ice-sheets, sea level and the climate over the last 21k years
- Supervisors: Dr. Feng He and Dr. Marianne Haseloff

Publications

- Hill, T., Hoffman, M. J., Flowers, G. E., & Bingham, D. (2026). Random forest parameterization of Antarctic subglacial hydrology for coupled ice-flow modelling. *EGUsphere preprint* [submitted 2026-01-29]. doi: [10.5194/egusphere-2026-343](https://doi.org/10.5194/egusphere-2026-343)
- Hill, T., Flowers G. E., Bingham, D., & Hoffman, M. J. (2025b) Emulator-based Bayesian calibration of a subglacial drainage model. *Annals of Glaciology*, 66, e22. doi:[10.1017/aog.2025.10016](https://doi.org/10.1017/aog.2025.10016).
- Hill, T., Bingham, D., Flowers, G. E., & Hoffman, M. J. (2025a) Computationally efficient subglacial drainage modelling using Gaussian Process emulators: GlaDS-GP v1.0. *Geoscientific Model Development*, 18, 4045-4074. doi: [10.5194/gmd-18-4045-2025](https://doi.org/10.5194/gmd-18-4045-2025)
- Killingbeck, S. F., Rutishauser, A., Unsworth, M. J., Dubnick, A., Criscitiello, A. S., Killingbeck, J., Dow, C. F., Hill T., Booth, Adam D., Main, B., & Brossier, E. (2024) Misidentified subglacial lake beneath the Devon Ice Cap, Canadian Arctic: a new interpretation from seismic and electromagnetic data. *The Cryosphere*, 18(8), 3699-3722. doi: [10.5194/tc-18-3699-2024](https://doi.org/10.5194/tc-18-3699-2024)
- Hill, T., Flowers, G. E., Hoffman, M. J., Bingham, D., and Werder, M. A. (2024) Improved representation of laminar and turbulent sheet flow in subglacial drainage models. *Journal of Glaciology*, 70, e24. doi: [10.1017/jog.2023.103](https://doi.org/10.1017/jog.2023.103) [2024 IGS Graham Cogley student paper award]
- Hill, T. and Dow, C. F. (2023) The impact of surface melt rate and catchment characteristics on Greenland Ice Sheet moulin inputs. *The Cryosphere*, 17, 2607–2624 doi: [10.5194/tc-17-2607-2023](https://doi.org/10.5194/tc-17-2607-2023)

- Hill, T.** and Dow, C. F. (2021) Modeling the Dynamics of Supraglacial Rivers and Distributed Meltwater Flow with the Subaerial Drainage System (SaDS) Model. *Journal of Geophysical Research: Earth Surface*, 126, e2021JF006309. doi: [10.1029/2021JF006309](https://doi.org/10.1029/2021JF006309)
- Hill, T.**, Dow, C. F., Bash, E. A., and Copland, L. (2021) Application of an improved surface energy balance model to two large valley glaciers in the St. Elias Mountains, Yukon. *Journal of Glaciology*, 67(262), 297-312. doi: [10.1017/jog.2020.106](https://doi.org/10.1017/jog.2020.106)
- Nassar, R., Mastrogiacomio, J. P., Bateman-Hemphill, W., McCracken, C., MacDonald, C. G., **Hill, T.**, O'Dell, C., Kiel, M., & Crisp, D. (2021). Advances in quantifying power plant CO₂ emissions with OCO-2. *Remote Sensing of Environment*, 264, 112579. doi: [10.1016/j.rse.2021.112579](https://doi.org/10.1016/j.rse.2021.112579)
- Hill, T.**, and Nassar, R. (2019) Pixel size and revisit rate requirements for monitoring power plant CO₂ emissions from space. *Remote Sensing*, 11(13): 1608. doi: [10.3390/rs11131608](https://doi.org/10.3390/rs11131608)
- Nassar, R., **Hill, T. G.**, McLinden, C., Wunch, D., Jones, D., and Crisp, D. (2017) Quantifying CO₂ emissions from individual power plants from space. *Geophysical Research Letters*, 44(19), 10045-10053. doi: [10.1002/2017GL074702](https://doi.org/10.1002/2017GL074702)

Funding and selected awards

2026/9 - 2028/9	Canada Postdoctoral Research Award, <i>NSERC Canada</i>	\$140,000
2024	IGS Graham Cogley Award	\$500
2024/8	Best oral presentation, <i>IGS model verification symposium</i>	-
2024/4	Outstanding Reviewer Award, <i>EGU/The Cryosphere</i>	-
2021/9 - 2024/9	Provost Prize of Distinction, <i>Simon Fraser University</i>	\$15,000
2021/9 - 2024/9	Canada Graduate Scholarship Doctoral, <i>NSERC Canada</i>	\$105,000
2021/7	Outstanding Teaching Assistant Award, <i>UWaterloo</i>	\$500
2020/9	Joseph Wai-Hung Liu Graduate Scholarship, <i>UWaterloo</i>	\$5,000
2020/9	Ontario Graduate Scholarship, <i>Government of Ontario</i>	\$15,000
2019/9	Canadian Graduate Scholarship Masters, <i>NSERC Canada</i>	\$17,500

Conference abstracts & presentations

[Invited]

- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2026) Subglacial drainage modelling: from fundamental physics to machine-learning emulation. IGS global seminar series, Online.
- Hill, T.**, Flowers, G., Hoffman, M., and Bingham, D. (2025) Emulator-based subglacial drainage modelling. Online seminar, Mathematics on Ice forum.
- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2025) Subglacial drainage modelling and Bayesian calibration using Gaussian Process emulators. Oral presentation at the 2025 EGU General Assembly, Vienna, Austria, EGU25-3771. <https://doi.org/10.5194/egusphere-egu25-3771>

[Contributed]

- Hill, T.**, Flowers, G., Hoffman, M., and Bingham, D. (2025) Random-Forest Parameterization of Antarctic Subglacial Hydrology. Poster presentation at the 2025 WAIS Workshop, Coupeville, USA.
- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2025) Subglacial drainage modelling and Bayesian calibration using Gaussian Process emulators. Seminar at SFU Earth Sciences Research Day, Vancouver, Canada.
- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2024) Gaussian process emulation of subglacial drainage. Oral presentation at the International Symposium on Verification and Validation of Cryospheric Models, Newcastle Upon Tyne, UK. **[Best Oral Presentation Award]**
- Hill, T.**, Flowers, G., Bingham, D., Hoffman, M., and Werder, M. (2023) Improved representation of laminar and turbulent sheet flow in subglacial drainage models. Oral presentation at the AGU Fall Meeting, San Francisco, USA, C51B-08.
- Hill, T.**, Flowers, G., Bingham, D., Hoffman, M., and Werder, M. (2023) Laminar and turbulent subglacial water flow. Oral presentation at the Annual Meeting of the Northwest Glaciologists, Seattle, USA.
- Killingbeck, S., Killingbeck, J., **Hill, T.**, Main, B., Brossier, E., Unsworth, M., Dow, C., Dubnick, A., Criscitiello, A., and Rutishauser, A. (2022) Combined Geophysical Techniques Constrain the Devon Ice Cap, Canadian High Arctic, Subglacial Environment. Oral presentation at AGU Fall Meeting, Chicago, USA, C42B-05.
- Hill, T.**, Flowers, G., and Bingham, D. (2022) Gaussian process emulation of subglacial drainage. Oral presentation at Annual Meeting of the Northwest Glaciologists, Moscow, USA.
- Siu, K., Dow, C. F., Morlighem, M., McCormack, F., and **Hill, T.** (2022) Modelling Subglacial Hydrology under Future Climate Scenarios in Wilkes Subglacial Basin, Antarctica. Oral presentation at EGU General Assembly 2022, Vienna, Austria, EGU22-424.
- Hill, T.**, and Dow, C. F. (2021) Modeling the Dynamics of Supraglacial Rivers and Distributed Meltwater Flow on the Greenland Ice Sheet With the Subaerial Drainage System (SaDS) Model. Poster presentation at AGU Fall Meeting, New Orleans, USA.
- Hill, T.**, and Dow, C. F. (2021) Modelling glacier melt rates and surface hydrology at the basin scale. Poster presentation at CGU Student Conference, Online.

Teaching

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| 2025/1 - 2025/4 | Teaching Assistant , Simon Fraser University
<i>EASC 305: Quantitative Methods for Earth Science</i> <ul style="list-style-type: none">• Develop, lead & grade weekly computer lab assignments |
| 2024/10 | Guest Lecturer , Simon Fraser University
<i>EASC 600: Introduction to graduate studies</i> : Intro to software version control <ul style="list-style-type: none">• Benefits and best practices around software version control• Interactive demonstration using GitHub online GUI |
| 2023/1 - 2023-2 | Accessible Learning Lab Aid , Simon Fraser University
<i>EASC 314: Principles of Glaciology</i> |

- One-on-one weekly computer lab support
- 2020/9 - 2020/12 **Teaching Assistant**, University of Waterloo
Math 227: Calculus 3 for Honours Physics
- Weekly office hours and grading

Student mentoring

- 2026/9 - 2027/4 **Supervisor**, Brett Nuthals, Undergraduate Research Assistant
Sea-level coupling of idealized marine ice sheets
- 2023/4 - 2023/8 **Supervisor**, Orion Vllasi, Undergraduate Research Assistant
Subglacial hydrology modelling with ISSM

Service

[Volunteer organizations]

- 2025/10 - 2026/10 **EX-Officio**, [APECS Canada](#)
- Ensure continuity and transfer of institutional knowledge
- 2023/10 - 2025/10 **Vice-Chair Administration & Finance**, [APECS Canada](#)
- Initiated [pro deal program](#) available to >500 Canadian polar ECRs
 - Chair of 2024 [Mentor Award](#) committee and 2025 member-at-large
- 2023 - 2025 **Graduate Student Society (GSS) Representative & Secretary**, Department of Earth Sciences Graduate Student Caucus

[Peer review]

- 2026 Geophysical Research Letters, GEUS Bulletin, Nature Communications
- 2025 Journal of Geophysical Research: Earth Surface, Nature Communications
 Journal of Glaciology, Journal of Artificial Intelligence Research, IASC cryosphere fellowship program
- 2024 Nature, Journal of Geophysical Research: Earth Surface,
 Geophysical Research Letters, IASC cryosphere fellowship program
- 2023 The Cryosphere [**EGU outstanding reviewer award**]
 Geophysical Research Letters
- 2022 Journal of Glaciology

Additional appointments

- 2023/8 **Field Assistant**, Geological Survey of Canada
- Monitoring alpine hazards, Joffre Lakes, BC
- Supervisor: Dr. Jeff Crompton
- 2019/5 - 2019/8 **Research Assistant**, ECCC Climate Research Division
- Estimating power plant CO₂ emissions from satellite data
 - Develop & maintain python/bash software for operational analysis
- Supervisor: Dr. Ray Nassar
- 2018/5 - 2018/8 **Undergrad Research Assistant**, University of Waterloo Applied Math

- Lake ice modelling with MITgcm using high-performance computing clusters

Supervisors: Dr. Marek Stastna and Dr. Kevin Lamb

2017/9 - 2017/12 **Undergrad Research Assistant**, Dr. K Resch, Institute for Quantum Computing

2016/9 - 2017/4 **Research Assistant**, Dr. R. Nassar, ECCC Climate Research Division

Software and products

Ice-sheet and Sea-level System Model (ISSM) contributions (github.com/ISSMTeam/ISSM)

- Contributed improvements to GlaDS subglacial drainage system component (Hill et al., 2024, J. Glaciol.) within one of the community's most popular ice-sheet models (>>500 citations)
- Contributed code has been used and tested at University of Alaska-Fairbanks and University of Waterloo, Canada

GPU-Accelerated Gaussian Processes (github.com/timghill/GPmodule-torch)

- Python and pytorch-based code for univariate Gaussian process modelling and MCMC sampling

Subglacial outburst flood model (github.com/SFUGG/outburst_flood_model)

- Pure-python finite-difference implementation of a canonical model of subglacial water flow
- Developed for teaching, used by G. Flowers at 2026 [International Summer School in Glaciology](#)

Subaerial Drainage System (SaDS) model

- Matlab 2D-planview and time-varying model for modelling water flow at the glacier surface
- Custom finite volume implementation (Hill & Dow, 2021 J. Geophys. Res. Earth Surf.)
- Used by University of Waterloo, Canada and Friedrich-Alexander University, Germany

Skills

HPC: SLURM, MPI, Apptainer (Digital Research Alliance of Canada, University of Wisconsin-Madison CHTC)

Languages: Python, pytorch, Fortran, C++, R, LaTeX

Community-developed models: MITgcm (Fortran/finite volume), ISSM (C++/finite element), PISM (C++/finite diff.), SELEN⁴ (fortran/spectral), GlaDS (matlab/C++/finite element), SEPIA (python/Bayesian inference)

Custom numerical method implementations: Finite volume (Hill & Dow, 2021 J. Geophys. Res. Earth Surf.), Finite element (ISSM), Multivariate Gaussian processes (Hill et al., 2025 Geosci. Mod. Devel.)